

# Scalability & Future Plan

|              |                                          |               |                          |
|--------------|------------------------------------------|---------------|--------------------------|
| Date         | 30 September 2026                        | Team ID       | 6ab3e57da238dc999eb3d44d |
| Project Name | EduGenie – AI-Powered Learning Assistant | Maximum Marks | As per template          |

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**GitHub:** <https://github.com/ricardovictor-28507/Google-Cloud-Generative-AI--A-S>

## Current limitations

| S.No | Limitation                             | Impact                                                       |
|------|----------------------------------------|--------------------------------------------------------------|
| 1    | No persistent learner profiles/history | Cannot track progress across sessions.                       |
| 2    | No authentication/authorization        | Not ready for multi-user production deployment.              |
| 3    | No database                            | No saved quizzes, notes or learning paths.                   |
| 4    | Gemini dependency for live generation  | External API availability and latency affect live responses. |
| 5    | No load benchmark yet                  | Production capacity is not quantified.                       |

## Scalability Plan

| Aspect            | Current State        | Proposed Upgrade                                                             |
|-------------------|----------------------|------------------------------------------------------------------------------|
| Application       | Single FastAPI app   | Containerized deployment with multiple workers.                              |
| AI service        | Single shared client | Async calls, retry/backoff and rate limiting.                                |
| Data              | Stateless            | PostgreSQL/managed storage for users and progress.                           |
| Security          | Environment API key  | Authentication, authorization, secrets manager.                              |
| Performance       | Local smoke test     | Load testing, caching where appropriate and monitoring.                      |
| Learning features | Five workflows       | PDF/image input, progress dashboard, multilingual support and teacher tools. |